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CLINICAL INVESTIGATION **Eye**

PROTON BEAM RADIOTHERAPY OF CHOROIDAL MELANOMA: THE LIVERPOOL-CLATTERBRIDGE EXPERIENCE

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Purpose: To report on outcomes after proton beam radiotherapy of choroidal melanoma using a 62-MeV cyclotron in patients considered unsuitable for other forms of conservative therapy.
Methods and Materials: A total of 349 patients with choroidal melanoma referred to the Liverpool Ocular Oncology Centre underwent proton beam radiotherapy at Clatterbridge Centre for Oncology (CCO) between January 1993 and December 2003. Four daily fractions of proton beam radiotherapy were delivered, with a total dose of 53.1 proton Gy, and with lateral and distal safety margins of 2.5 mm. Outcomes measured were local tumor recurrence; ocular conservation; vision; and metastatic death according to age, gender, eye, visual acuity, location of anterior and posterior tumor margins, quadrant, longest basal tumor dimension, tumor height, extraocular extension, and retinal invasion.
Results: The 5-year actuarial rates were 3.5% for local tumor recurrence, 9.4% for enucleation, 79.1% for conservation of vision of counting fingers or better, 61.1% for conservation of vision of 20/200 or better, 44.8% for conservation of vision of 20/40 or better, and 10.0% for death from metastasis.
Conclusion: Proton beam radiotherapy with a 62 MeV cyclotron achieves high rates of local tumor control and ocular conservation, with visual outcome depending on tumor size and location. © 2005 Elsevier Inc.

Uveal melanoma, Accelerated radiotherapy, Metastasis, Vision, Tumor control.

INTRODUCTION

Choroidal melanoma can cause visual handicap, loss of the eye, and metastatic disease, which is invariably fatal (1). For many years, the standard form of therapy was enucleation, but this has been replaced whenever possible by a variety of therapies aimed at destroying the primary tumor while conserving the eye, ideally with useful vision. Such "conservative" methods include: brachytherapy, using isotopes such as iodine-125 or ruthenium-106; proton beam radiotherapy; stereotactic radiotherapy; local resection, which can be transscleral or transretinal; and transpupillary thermotherapy (1).

Proton beam radiotherapy of uveal melanomas was first developed in Boston and is now performed in more than a dozen centers around the world (2). In Boston and some other centers, it is used for all patients; however, it is more expensive and time-consuming than brachytherapy and can also cause side effects in extraocular structures, such as eyelids, lacrimal gland, and the tear ducts. We therefore use proton beam radiotherapy selectively, when other methods are considered less likely to be successful (3). We mostly select proton beam radiotherapy for small, posterior tumors, because with such tumors we feel that it achieves local tumor control more reliably than brachytherapy while delivering smaller doses of radiation to the optic nerve and fovea. We also select proton beam radiotherapy when the tumor is too large for ruthenium plaque brachytherapy, if the patient is highly motivated to retain the eye, and if local resection is not possible.

Several centers have already reported their results with proton beam radiotherapy of uveal melanoma (4–10). However, our selection criteria are different. Furthermore, the Clatterbridge Centre for Oncology (CCO) proton beam is highly conformal in comparison to other units, and differences in treatment technique have evolved at different centers.

The aim of this study was to determine ocular outcomes such as visual acuity, local tumor control, ocular retention,

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